

Skill-4&5

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Step-1: Insert Customer Data

Customer 1:

```
{  
  "customer_id": 101,  
  "name": "Ravi Kumar",  
  "city": "Hyderabad",  
  "account_type": "Savings",  
  "balance": 85000,  
  "status": "Active"  
}
```

Customer 2

```
{  
  "customer_id": 102,  
  "name": "Priya Sharma",  
  "city": "Vijayawada",  
  "account_type": "Current",  
  "balance": 125000,  
  "status": "Active"  
}
```

Customer 3

```
{  
  "customer_id": 103,
```

```
"name": "Arjun Reddy",  
"city": "Hyderabad",  
"account_type": "Savings",  
"balance": 45000,  
"status": "Active"  
}
```

Customer 4

```
{  
  "customer_id": 104,  
  "name": "Sneha Rao",  
  "city": "Warangal",  
  "account_type": "Savings",  
  "balance": 210000,  
  "status": "Active"  
}
```

Customer 5

```
{  
  "customer_id": 105,  
  "name": "Kiran Kumar",  
  "city": "Guntur",  
  "account_type": "Current",  
  "balance": 65000,  
  "status": "Inactive"  
}
```

Customer 6

```
{  
  "customer_id": 106,  
  "name": "Anjali Singh",  
  "city": "Hyderabad",  
  "account_type": "Savings",  
  "balance": 175000,  
  "status": "Active"  
}
```

Customer 7

```
{  
  "customer_id": 107,  
  "name": "Rahul Verma",  
  "city": "Bengaluru",  
  "account_type": "Current",  
  "balance": 95000,  
  "status": "Active"  
}
```

Customer 8

```
{  
  "customer_id": 108,  
  "name": "Meena Devi",  
  "city": "Chennai",  
  "account_type": "Savings",  
  "balance": 320000,  
  "status": "Active"
```

```
}
```

Step-2: Insert Transaction Data

Transaction 1

```
{  
  "transaction_id": 1001,  
  "customer_id": 101,  
  "type": "Deposit",  
  "amount": 25000,  
  "mode": "UPI",  
  "date": "2026-08-01",  
  "status": "Success"  
}
```

Transaction 2

```
{  
  "transaction_id": 1002,  
  "customer_id": 101,  
  "type": "Withdrawal",  
  "amount": 10000,  
  "mode": "ATM",  
  "date": "2026-08-02",  
  "status": "Success"  
}
```

Transaction 3

```
{  
  "transaction_id": 1003,
```

```
"customer_id": 102,  
"type": "Deposit",  
"amount": 50000,  
"mode": "NEFT",  
"date": "2026-08-02",  
"status": "Success"  
}
```

Transaction 4

```
{  
  "transaction_id": 1004,  
  "customer_id": 102,  
  "type": "Withdrawal",  
  "amount": 15000,  
  "mode": "ATM",  
  "date": "2026-08-03",  
  "status": "Success"  
}
```

Transaction 5

```
{  
  "transaction_id": 1005,  
  "customer_id": 103,  
  "type": "Deposit",  
  "amount": 30000,  
  "mode": "UPI",  
  "date": "2026-08-03",
```

```
"status": "Success"
}
```

Transaction 6

```
{
  "transaction_id": 1006,
  "customer_id": 103,
  "type": "Withdrawal",
  "amount": 5000,
  "mode": "ATM",
  "date": "2026-08-04",
  "status": "Success"
}
```

Transaction 7

```
{
  "transaction_id": 1007,
  "customer_id": 104,
  "type": "Deposit",
  "amount": 100000,
  "mode": "NEFT",
  "date": "2026-08-04",
  "status": "Success"
}
```

Transaction 8

```
{
  "transaction_id": 1008,
```

```
"customer_id": 104,  
"type": "Withdrawal",  
"amount": 25000,  
"mode": "ATM",  
"date": "2026-08-05",  
"status": "Success"  
}
```

Transaction 9

```
{  
  "transaction_id": 1009,  
  "customer_id": 105,  
  "type": "Deposit",  
  "amount": 15000,  
  "mode": "UPI",  
  "date": "2026-08-05",  
  "status": "Success"  
}
```

Transaction 10

```
{  
  "transaction_id": 1010,  
  "customer_id": 106,  
  "type": "Deposit",  
  "amount": 75000,  
  "mode": "NEFT",  
  "date": "2026-08-06",
```

```
"status": "Success"
}
```

Transaction 11

```
{
  "transaction_id": 1011,
  "customer_id": 106,
  "type": "Withdrawal",
  "amount": 20000,
  "mode": "ATM",
  "date": "2026-08-06",
  "status": "Success"
}
```

Transaction 12

```
{
  "transaction_id": 1012,
  "customer_id": 107,
  "type": "Deposit",
  "amount": 40000,
  "mode": "IMPS",
  "date": "2026-08-07",
  "status": "Success"
}
```

Transaction 13

```
{
  "transaction_id": 1013,
```



```
"customer_id": 107,  
"type": "Withdrawal",  
"amount": 12000,  
"mode": "ATM",  
"date": "2026-08-07",  
"status": "Success"  
}
```

Transaction 14

```
{  
  "transaction_id": 1014,  
  "customer_id": 108,  
  "type": "Deposit",  
  "amount": 150000,  
  "mode": "NEFT",  
  "date": "2026-08-08",  
  "status": "Success"  
}
```

Transaction 15

```
{  
  "transaction_id": 1015,  
  "customer_id": 108,  
  "type": "Withdrawal",  
  "amount": 30000,  
  "mode": "ATM",  
  "date": "2026-08-08",
```

```
"status": "Success"
```

```
}
```

Step-3: Execute the Queries:

Query 1 — Display All Customers

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'clusters' list with 'Cluster0' selected. The main panel shows the 'customers' collection under 'bankdb'. The query bar is empty, and the results table displays 8 documents. The table has columns: _id, ObjectID, customer_id, name, city, account_type, and balance.

	_id	ObjectID	customer_id	name	city	account_type	balance
1	ObjectID('6a7c3c1b8f5bae...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	
2	ObjectID('6a7c3c4db8f5bae...')	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	
3	ObjectID('6a7c3c5fb8f5bae...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	
4	ObjectID('6a7c3c83b8f5bae...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000	
5	ObjectID('6a7c3c91b8f5bae...')	105	"Kiran Kumar"	"Guntur"	"Current"	65000	
6	ObjectID('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	
7	ObjectID('6a7c3cafb8f5bae...')	107	"Rahul Verma"	"Bengaluru"	"Current"	95000	
8	ObjectID('6a7c3ccdb8f5bae...')	108	"Meena Devi"	"Chennai"	"Savings"	320000	

Query 2 — Find Active Customers

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'clusters' list with 'Cluster0' selected. The main panel shows the 'customers' collection under 'bankdb'. The query bar contains the filter `{ "status": "Active" }`. The results table displays 7 documents. The table has columns: name, city, account_type, balance, and status.

	name	city	account_type	balance	status
1	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
3	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
4	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
5	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
6	"Rahul Verma"	"Bengaluru"	"Current"	95000	"Active"
7	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

Query 3 — Balance Greater Than 100000

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the cluster hierarchy: Cluster0, admin, bankdb, and customers. The main panel shows the 'customers' collection with a filter query: `{ "balance": { "$gt": 100000 } }`. The 'Documents' tab is active, showing a table of 4 documents. The table has columns: _id, ObjectId, customer_id, Int32, name, String, city, String, account_type, String, and balance. The data rows are:

	_id	ObjectId	customer_id	Int32	name	String	city	String	account_type	String	balance
1	ObjectId('6a7c3c4db8f5bae...')	102	"Priya Sharma"	"Vijayawada"	"Current"	125000					
2	ObjectId('6a7c3c83b8f5bae...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000					
3	ObjectId('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000					
4	ObjectId('6a7c3ccdb8f5bae...')	108	"Meena Devi"	"Chennai"	"Savings"	320000					

Query 4 — Hyderabad Customers

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the cluster hierarchy: Cluster0, admin, bankdb, and customers. The main panel shows the 'customers' collection with a filter query: `{ "city": "Hyderabad" }`. The 'Documents' tab is active, showing a table of 3 documents. The table has columns: _id, ObjectId, customer_id, Int32, name, String, city, String, account_type, String, and balance. The data rows are:

	_id	ObjectId	customer_id	Int32	name	String	city	String	account_type	String	balance
1	ObjectId('6a7c3c10b8f5bae...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000					
2	ObjectId('6a7c3c5fb8f5bae...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000					
3	ObjectId('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000					

Query 5 — Savings Account Customers

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the cluster hierarchy: Cluster0, admin, bankdb, and customers. The main panel shows the 'customers' collection with a filter query: `{ "account_type": "Savings" }`. The 'Documents' tab is active, showing a table of 5 documents. The table has columns: _id, ObjectId, customer_id, Int32, name, String, city, String, account_type, String, and balance. The data rows are:

	_id	ObjectId	customer_id	Int32	name	String	city	String	account_type	String	balance
1	ObjectId('6a7c3c10b8f5bae...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000					
2	ObjectId('6a7c3c5fb8f5bae...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000					
3	ObjectId('6a7c3c83b8f5bae...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000					
4	ObjectId('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000					
5	ObjectId('6a7c3ccdb8f5bae...')	108	"Meena Devi"	"Chennai"	"Savings"	320000					

Query 6 — Transactions Greater Than 20000

Organization: SIMRAN JANGID's Or... Project: Project 0

Data Explorer: My Queries, Data Modeling

CLUSTERS (1): Cluster0 (admin, bankdb, customers, transactions, local, sample_mflix)

Cluster0 > bankdb > transactions

Documents: 15 Aggregations Schema Indexes: 1 Validation

Query: `{ "amount": { "$gt": 20000 } }` [Generate query] [Explain] [Reset] [Find]

Buttons: ADD DATA, BULK, EXPORT CODE

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode S
1	ObjectId('6a7c3cebb8f5bae...')	1001	101	"Deposit"	25000	"UPI"
2	ObjectId('6a7c3d18b8f5bae...')	1003	102	"Deposit"	50000	"NEFT"
3	ObjectId('6a7c3d3db8f5bae...')	1005	103	"Deposit"	30000	"UPI"
4	ObjectId('6a7c3d57b8f5bae...')	1007	104	"Deposit"	100000	"NEFT"
5	ObjectId('6a7c3d65b8f5bae...')	1008	104	"Withdrawal"	25000	"ATM"
6	ObjectId('6a7c3d85b8f5bae...')	1010	106	"Deposit"	75000	"NEFT"
7	ObjectId('6a7c3da4b8f5bae...')	1012	107	"Deposit"	40000	"IMPS"

Query 7 — Find Deposits

Organization: SIMRAN JANGID's Or... Project: Project 0

Data Explorer: My Queries, Data Modeling

CLUSTERS (1): Cluster0 (admin, bankdb, customers, transactions, local, sample_mflix)

Cluster0 > bankdb > transactions

Documents: 15 Aggregations Schema Indexes: 1 Validation

Query: `{ "type": "Deposit" }` [Generate query] [Explain] [Reset] [Find]

Buttons: ADD DATA, BULK, EXPORT CODE

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode S
1	ObjectId('6a7c3cebb8f5bae...')	1001	101	"Deposit"	25000	"UPI"
2	ObjectId('6a7c3d18b8f5bae...')	1003	101	"Deposit"	50000	"NEFT"
3	ObjectId('6a7c3d3db8f5bae...')	1005	103	"Deposit"	30000	"UPI"
4	ObjectId('6a7c3d57b8f5bae...')	1007	104	"Deposit"	100000	"NEFT"
5	ObjectId('6a7c3d74b8f5bae...')	1009	105	"Deposit"	15000	"UPI"
6	ObjectId('6a7c3d85b8f5bae...')	1010	106	"Deposit"	75000	"NEFT"
7	ObjectId('6a7c3da4b8f5bae...')	1012	107	"Deposit"	40000	"IMPS"
8	ObjectId('6a7c3dbfb8f5bae...')	1014	108	"Deposit"	150000	"NEFT"

Query 8 — Find ATM Transactions

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- customers
- transactions**
- local
- sample_mflix

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find

ADD DATA BULK EXPORT CODE

25 1 - 7 of 7

		transaction_id Int32	customer_id Int32	type String	amount Int32	mode String	da
1	5bae...	1002	101	"Withdrawal"	10000	"ATM"	"2"
2	5bae...	1004	102	"Withdrawal"	15000	"ATM"	"2"
3	5bae...	1006	103	"Withdrawal"	5000	"ATM"	"2"
4	5bae...	1008	104	"Withdrawal"	25000	"ATM"	"2"
5	5bae...	1011	106	"Withdrawal"	20000	"ATM"	"2"
6	5bae...	1013	107	"Withdrawal"	12000	"ATM"	"2"
7	5bae...	1015	108	"Withdrawal"	30000	"ATM"	"2"

Query 9 — Deposits Greater Than 50000

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- customers
- transactions**
- local
- sample_mflix

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find

ADD DATA BULK EXPORT CODE

25 1 - 3 of 3

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode Str
1	ObjectId('6a7c3d57b8f5bae...')	1007	104	"Deposit"	100000	"NEFT"
2	ObjectId('6a7c3d85b8f5bae...')	1010	106	"Deposit"	75000	"NEFT"
3	ObjectId('6a7c3dbfb8f5bae...')	1014	108	"Deposit"	150000	"NEFT"

Query 10 — Hyderabad Customers with Balance > 100000

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- customers**
- transactions
- local

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find

ADD DATA BULK EXPORT CODE

25 1 - 1 of 1

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance
1	ObjectId('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000

Query 11 — Customers From Multiple Cities

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- local
- sample_mflix

Cluster0 > bankdb > customers

Documents 8 Aggregations Schema Indexes 1 Validation

`{ "city": { "$in": [ "Hyderabad", "Vijayawada", "Warangal" ] }}`

Generate query Explain Reset

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32
1	ObjectId('6a7c3c10b8f5bae...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000
2	ObjectId('6a7c3c4db8f5bae...')	102	"Priya Sharma"	"Vijayawada"	"Current"	125000
3	ObjectId('6a7c3c5fb8f5bae...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000
4	ObjectId('6a7c3c83b8f5bae...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000
5	ObjectId('6a7c3c9fb8f5bae...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000

Step-4: Aggregation Queries

Query 12 — Sort Customers by Balance

Stage 1: Sort ENABLED ADD STAGE

STAGE INPUT Sample of 6 documents

OPTIONS

Sort Open docs

```
1 * /**
2 * Provide any number of field/order pairs.
3 */
4 {
5   "balance": -1
6 }
```

STAGE OUTPUT Sample of 6 documents

OPTIONS

Sample of 6 documents

1 ObjectId('6a7c3c10b8f5bae...') customer_id: 101 name: "Ravi Kumar" city: "Hyderabad" account_type: "Savings" balance: 85000 status: "Active"

2 ObjectId('6a7c3c4db8f5bae...') customer_id: 102 name: "Priya Sharma" city: "Vijayawada" account_type: "Current" balance: 125000 status: "Active"

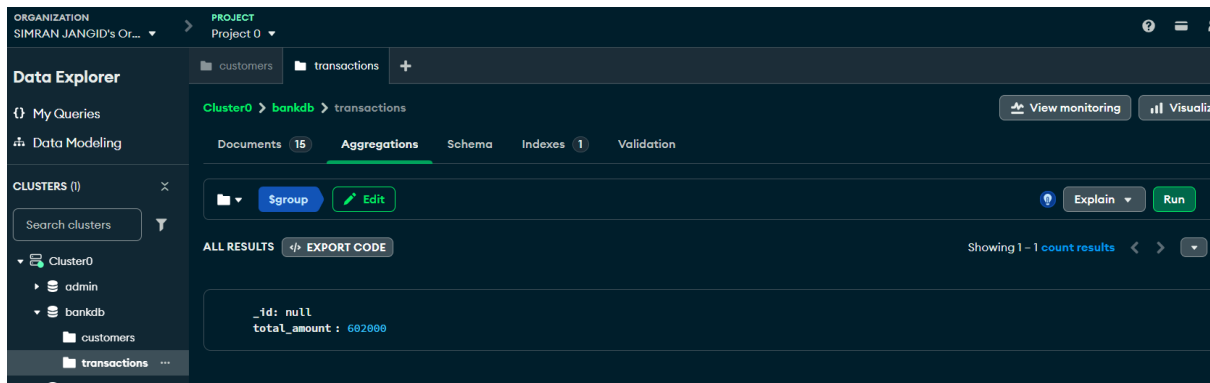
3 ObjectId('6a7c3c5fb8f5bae...') customer_id: 103 name: "Arjun Reddy" city: "Hyderabad" account_type: "Savings" balance: 45000 status: "Active"

4 ObjectId('6a7c3c83b8f5bae...') customer_id: 104 name: "Sneha Rao" city: "Warangal" account_type: "Savings" balance: 210000 status: "Active"

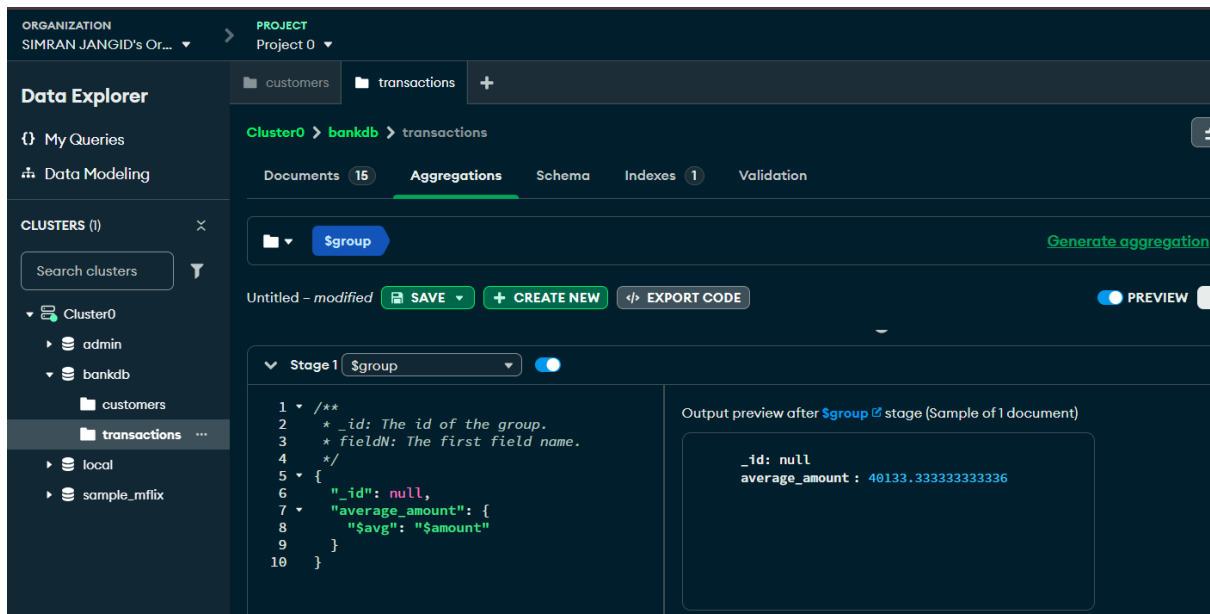
5 ObjectId('6a7c3c9fb8f5bae...') customer_id: 106 name: "Anjali Singh" city: "Hyderabad" account_type: "Savings" balance: 175000 status: "Active"

6 ObjectId('6a7c3c4db8f5bae...') customer_id: 102 name: "Priya Sharma" city: "Vijayawada" account_type: "Current" balance: 125000 status: "Active"

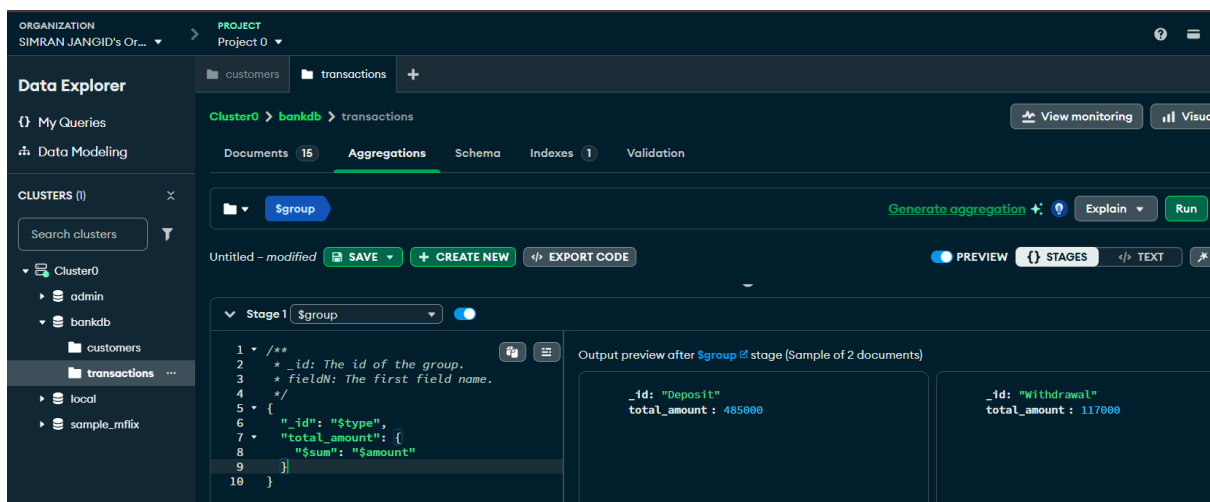
Query 13 — Total Transaction Amount



Query 14 — Average Transaction Amount



Query 15 — Total Amount by Transaction Type



Query 16 — Number of Transactions per Customer

The screenshot displays the MongoDB Compass interface during an aggregation process. At the top, there are tabs for '\$group' and '\$sort'. Below the tabs, there's a section for 'Untitled - modified' with buttons for 'SAVE', 'CREATE NEW', and 'EXPORT CODE'. A 'PREVIEW' toggle is also visible.

Stage 1: \$group

```

1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   "_id": "$customer_id",
7   "transaction_count": {
8     "$sum": 1
9   }
10 }
```

Output preview after \$group stage (Sample of 8 documents)

_id	transaction_count
106	2
105	1

Stage 2: \$sort

```

1 /**
2  * Provide any number of field/order pair(s)
3  */
4 {
5   "transaction_count": -1
6 }
```

Output preview after \$sort stage (Sample of 8 documents)

_id	transaction_count
102	2
101	2